

DoS & DDoS: Volumetric, Protocol, Application

The three classes

Class	Attack	Examples
Volumetric	Saturate bandwidth	UDP flood, ICMP flood, amplification (DNS/NTP), huge DNS queries
Protocol/stateful	Exhaust server states	SYN flood (half-open), connection exhaustion, slowloris, ping-of-death
Application	Exhaust app resources	HTTP flood, expensive-query floods, PDF bombs, login storms, zip bombs

Classic flavors in one place

```
# SYN flood
# tool: hping3 -S -p 80 --flood target (lab only)
# DNS amplification
# small query -> big response to spoofed victim IP
# Slowloris
# open many HTTP conns, drip partial requests -> exhaust Apache's threads
# HTTP flood
# many GET / requests via botnet or IP rotation
```

Why amplification is worst

Attacker sends 60-byte DNS query with **spoofed victim IP**; resolver answers with 4KB+. Ratio ~70:1. A small botnet of spoofers through open resolvers = a huge downstream flood.

The layered defense

```
Edge: CDN (Cloudflare/Akamai) absorbs volumetric
|-> network: DDoS scrubbing centers
|-> firewall: rate limits, SYN-proxy, stateful connection cap
```

```
|-> WAF: filter HTTP floods, CAPTCHA for humans  
|-> app: slow-blow code, async, load-balanced, autoscaled  
|-> capacity: baseline headroom + autoscale
```

Detection & response favorites

Baseline traffic (bytes/sec, connections/sec, PPS) → anomaly alerts
Blackhole/redirect ISPs for the worst volumetric
Source-hash authentication for SYN floods (SYN cookies)
Human-intent detection for app floods (behavior patterns)

DoS is a war of economics: the attacker spends pennies against your peak capacity. That's why defense is capacity + filtering + cost-shifting to CDN – never a single box.

[← Module start](#)

[Botnets & Amplification Attack...](#)